

Anika Tahsin Meem

Curriculum Vitae

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Bashundhara R/A - Dhaka, Bangladesh

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WORK EXPERIENCE

Research Assistant	North South University, Dhaka, Bangladesh
	Advisor: Md. Mamun Molla Research Group: Center of Applied and Computational Science at NSU
April 2025 - Present	Numerical simulation of double diffusive convective heat and mass transfer, natural convection of Power-law fluids in two concentric cylinders with exothermic chemical reaction - A two-dimensional model is developed using generalized curvilinear coordinates, and the governing equations for momentum, energy, and mass transport are solved numerically via the implicit finite volume method. - This work provides practical insights for improving the performance of energy systems, catalytic reactors, and process industries involving non-Newtonian and reactive fluids.
Aug 2024 - March 2024	Machine Learning-Driven CFD: Insights into Natural Convection and Non-Newtonian Fluid Dynamics Around Heated Elliptic cylinders <i>(Accepted to Case Studies in Thermal Engineering)</i> - Co-authored the research proposal for fund management, collaborating with corresponding professor to secure funding for this project - Optimized engineering systems like heat exchangers and reactors, improving heat transfer, mass transport, and flow stability in non-Newtonian fluids. - Applied surrogate machine learning for correlation and providing quicker and more accurate solutions for industrial, renewable energy, and thermal management applications.
Nov 2023 - July 2024	Bio-Magnetic Field Effects on Pulsatile Non-Newtonian Blood Flow with Gold Nanoparticles in a Bifurcated Artery in Presence of Aneurysm <i>(Accepted to Physics Scripta)</i> - Developed mathematical models of human blood flow incorporating nanofluids. - Utilized the finite element method (FEM) to analyze bio-convection and heat transfer in aneurysmal arteries with gold nanoparticles under a magnetic field. - Contributed to research optimizing therapeutic strategies in cardiovascular medicine.
IT Coordinator	International Study Destination (ISD), Dhaka, Bangladesh
Jan 2023 - Oct 2023	- Front-end Development • Illustration • Poster Design • SOP review - Operations & Marketing • Meeting Conduct • Video Editing

**Software
Development Intern**
(Remote)

Oct 2021 - Feb 2023

GaoTek Inc., New York

- Team Leader
- Website Maintaining
- Digital Marketing

EDUCATION

2022

Bachelor of Science in Computer Science and Engineering

North South University, Dhaka, Bangladesh

CGPA: 3.29/4.0

RESEARCH INTERESTS

- Data Science,
- Medical Image Processing,
- Computer Vision,
- Computational Psychology,
- Time-Series.
- CFD

BACHELOR THESIS

**Developed an AI-
powered healthcare
platform**

- Developed a multi-disease detection system using deep learning and AI techniques.
- Diagnoses six diseases with multiple DL models 88%-99% accuracy—pneumonia, malaria, melanoma, brain tumor, breast cancer, and lung cancer.
- Integrated a Django-based web platform with telemedicine features.
- Offers video consultations and online prescriptions for accessible, affordable healthcare.

DIRECTED THESIS

**Predicting
Cryptocurrency
Price Drops Using
Time-Series
Analysis and Deep
Learning**

- Developed a deep learning stacking ensemble model for cryptocurrency price drop prediction.
- Utilized CNN, LSTM, BiLSTM, and GRU to analyze 20 key market parameters.
- Achieved high RMSE score between 0.0089 and 1.3229, outperforming existing models.
- Helps investors make proactive, informed decisions in a volatile market.

RESEARCH WORK

**A Cost-Effective
ML and DL
Solutions for
Predicting Arsenic
Contamination in
Groundwater**
(ongoing)

- Developed a cost-effective arsenic contamination prediction model using several ML/DL algorithms (RF, DT, XGBoost, CNN, GAN).
- Used accessible data like latitude, well depth, and lithology to minimize reliance on post-digging tests.
- With six parameters enhancing detection accuracy around 93.75% using CGAN.
- Built a Django web app and XAI for instant arsenic risk feedback based on user-input parameters.

PROJECT EXPERIENCE

2019	Junior Deign Project: Covid-19 detection using Deep learning and AI
2019	Software Development: Web-Based Lost and Found System.
2020	Deep-Learning: Based on MRI Brain Tumor Classification Using CNN and Autoencoder.
2020	DBMS: Direct From My Kitchen (Food App).
2021	Machine Learning: AI Fake News Detection Using Naive Bayes.
2021	Machine Learning: Identifying Socially Isolated People Using LSNS6.

JOURNAL PAPER

Published	Meem, Anika Tahsin, Mohammad Monirujjaman Khan, Mehedi Masud, and Sultan Aljahdali. "Prediction of Covid-19 Based on Chest X-Ray Images Using Deep Learning with CNN." Computer Systems Science & Engineering 41, no. 3 (2022).
Published	Akter, S., Meem, A. T., Akter, H., Molla, M. M., Souai, S., & Islam, S. (2025). Thermophoretic effects on thermosolutal natural convection in concentric cylindrical systems. Chinese Journal of Physics.
Published	Meem AT, Hossain MZ, Molla MM, Saha SC, Saha G. Biomagnetic field effects on pulsatile non-Newtonian blood flow with gold nanoparticles in a bifurcated artery in the presence of aneurysm. Pramana. 2025 Sep;99(3):1-37.
Published	Meem, Anika Tahsin, et al. "Data-driven insights of flow over heated elliptic cylinders: Machine learning and CFD perspectives on non-Newtonian forced convection." Case Studies in Thermal Engineering (2025): 106753.
Published	<u>Meem, Anika Tahsin, et al. "OVT-Net: Semantic Segmentation of Gastrointestinal Cancer Using an Optimized Vision Transformer Model with Explainable AI" Clinical eHealth (2025).</u>

CERTIFIED COURSE

Pantech Solutions	Master Class of Machine Learning and Artificial Intelligence.
Open Weaver	Build an AI Fake News Detection.
Udemy	Google Data Studio A-Z for Data Visualization and Dashboards.
Udemy	Professional Adobe Photoshop CC Course with Advance Training

TECHNICAL SKILLS

Programming & Web Development Languages	Python, C, HTML, CSS, Java, JavaScript, SQL
Frameworks	Django, GUI, FLASK
Tools	Tensorflow/Keras, OpenCV, NLP, Latex, Git, VSCode, Google Colab
Software	Canva, Power BI, Adobe Illustration, Figma, Matlab, COMSOL, Tecplot

SOFT SKILLS

- Collaborative Team Player, • Adaptive Contributor, • Detail-Oriented,
- Insightful Data Navigator, • Compelling Storyteller, • Time-management.

SCHOLARSHIPS

Tuition Waiver	North South University	<i>Course: Mat112, Eng102</i>
General Scholarship	Junior School Certificate - 2011	

LANGUAGES

English	Professional	<i>Duolingo: 115 (nblt 95)</i>
Bangla	Native	

OTHER ACTIVITIES

Assisted in Peer Review of Published Research	Assisted my research supervisor in reviewing multiple journal articles, providing insights on methodologies and key findings. • International Journal for Numerical Methods in Biomedical Engineering • Journal of Nanomaterials, Nanoengineering and Nanosystems • Multidiscipline Modeling in Materials and Structures • Modern Physics Letters B
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Interests	Poetry, Painting.
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VOLUNTEERING	North South University 26th Convocation (Volunteer for Marshal) • Assisted in managing convocation proceedings, guiding graduates and guests, and ensuring smooth coordination during the ceremony.
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LINKS

Email 2	meemanika70@gmail.com
ResearchGate	<u>Anika-Meem</u>
Instagram	<u>dear_anomaly</u>

REFERENCES

- Md. Mamun Molla**
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- Mohammad Monirujjaman Khan**
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